

Name: _____

$\triangle ABC \sim \triangle DEF$. Use the similarity statement for 1-2.

1. Name the three pairs of congruent angles and the three equal side ratios.
2. $AB = 28$ and $DE = 21$. What is the scale factor of $\triangle ABC$ to $\triangle DEF$? Of $\triangle DEF$ to $\triangle ABC$?

Similar or not? Show the ratio check.

3. Quadrilaterals with corresponding angles congruent and sides 6, 8, 10, 12 and 9, 12, 15, 18.
4. Rectangles 15×9 and 25×15 .
5. Rectangles 12×8 and 18×10 .

Solve.

6. $\triangle GHJ \sim \triangle MNP$, $GH = 28$, $MN = 21$, $HJ = 4x$, and $NP = 27$. Find x and HJ .
7. $ABCD \sim WXYZ$, $AB = 45$, $WX = 27$, $DA = 40$, and $ZW = 3x + 3$. Find x .

Perimeters.

8. Two similar triangles have scale factor $5/8$. The smaller has perimeter 40. Find the larger perimeter.
9. A photo measures 4 in. $\times$ 6 in. A similar print is 10 in. wide (matching the 4 in. side). Find its length and perimeter.
10. Are all squares similar? Are all rectangles? Explain the difference.